

A Report for Energy Management with AI-Methods



“Simulation and Optimization of Microgrid Electricity and Heat supply”

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Executive Summary

Study Overview

This study presents the design, simulation, and optimization of a microgrid system intended to supply electricity and heat to a small village while minimizing carbon emissions. Solar photovoltaic (PV) energy was modeled as the primary energy source. Multiple energy conversion technologies were integrated to transform solar energy into usable electricity and heat. Battery storage and a Power-to-Gas (PtG) system were incorporated to meet daily and seasonal electricity demand during periods of low or absent solar generation. A diesel generator set was included as a backup source to ensure system reliability during component failure or extreme peak demand conditions. Excess PV electricity is exported to the external grid in order to prevent curtailment.

For the heating sector, a groundwater heat pump was selected to supply the average heat demand, while a natural gas boiler was used to cover peak thermal loads. Thermal energy storage was designed based on the assumed annual operating hours of the heat pump and the amount of excess heat generated during periods of low demand. The storage system is capable of functioning as both daily and seasonal thermal storage, thereby eliminating the need for a separate seasonal heat storage system.

After selecting the system components, the energy system was modeled, simulated, and optimized using Python within an open-source JupyterHub environment. System operating hours, downtime, supply-demand balances, and the Levelized Cost of Energy (LCOE) were calculated based on the optimization results. The ecological assessment demonstrates significant reductions in primary energy consumption and CO₂ emissions compared with conventional supply scenarios.

Key Results Table

In the table below, we have provided the impact our energy model has created on Primary energy consumption per year and CO₂ emission per year in comparison with two other scenarios. In scenario 2, we have considered that electricity is being supplied from the grid, while heat demand is fulfilled by natural gas boiler. For scenario 3, the source of electricity has kept as electricity grid, while for heat demand, two converters are considered as our energy model in which electricity for heat pump is provided by electricity grid while fuel for boiler is natural gas. The comparison shows that by adopting this energy system, we can lower down primary energy consumption by 85% and 82% in comparison to scenario 2 and scenario 3., while the reduction in CO₂ emission in both cases is around 87%.

Cases	Primary Energy consumption (kWh/a)	CO ₂ Emission (kg CO ₂ eq)
Scenario 1	207028.81	45169.92
Scenario 2	1299860.46	340807.22
Scenario 3	1098282.34	328829.96

Table 1: Ecological comparison

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1 Introduction

Nowadays climate change has been a bigger issue worldwide. As a result of use of high carbon emitting energy fuels, World is facing many issues such as Higher AQI, Water scarcity, Average global temperature rise and many more. To find a solution to this global problem, major countries of the world under the Paris Agreement has decided **“to limit global warming to well below 2°C (3.6°F) above pre-industrial levels, while pursuing efforts to keep it to 1.5°C (2.7°F), significantly reducing climate change risks and impacts [1].”** Deadline to achieve this target is varied worldwide and Germany has decided to achieve this goal by the end of the year 2045(Climature change act,2021).

To meet its carbon neutrality goal, Germany has decided to target energy industries to get zero carbon emitting industry by adopting renewable energy and supplying green electricity and heat across the country. As a result, municipalities need to submit their energy road map to achieve this goal. To achieve this goal, we tried to simulate and optimize one microgrid, which can fulfill the electricity demand and heat demand of a small village, using solar energy as primary source of energy and other energy converters and storages to meet the demand in the absence of solar energy using Python Programming into open source JupyterHub.

Objective: To simulate and optimize microgrid using python programming having Solar energy as main source of energy with various energy converters and storages with lowest LCOE to fulfill users Electricity and Heat demand for a year.

System components being considered in the simulation are:

- PV system (Main source of energy)
- Battery storage, which should fulfill daily electricity need in absence of solar energy
- PtoG storage system, which converts electricity into hydrogen and hydrogen to the electricity when demanded (seasonal electricity storage)
- Heat pump (To meet average heat demand)
- Natural gas boiler (To meet peak heat demand)
- Heat storage (Can function as daily and seasonal heat storage)
- Diesel genset (Backup electricity source)

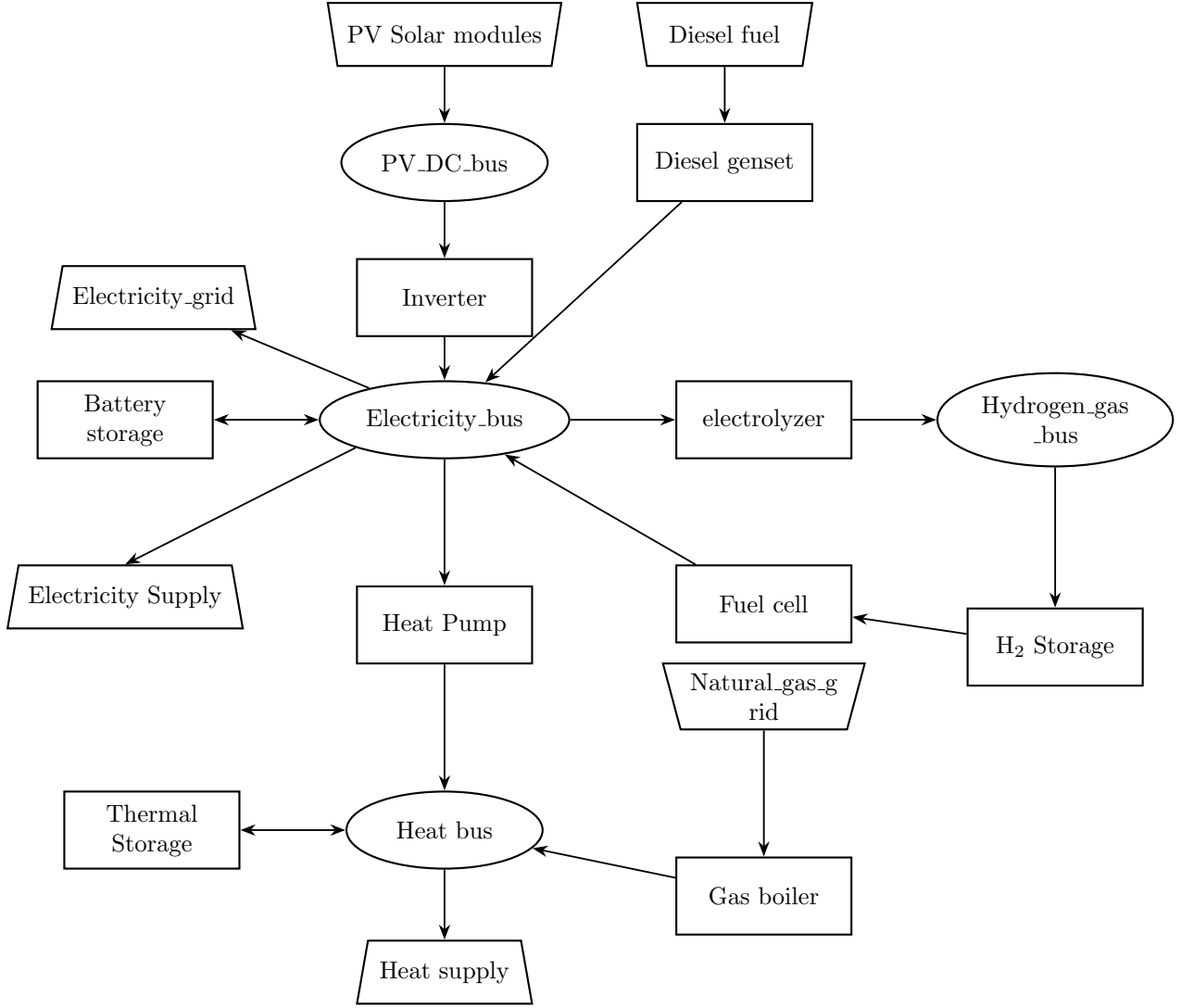


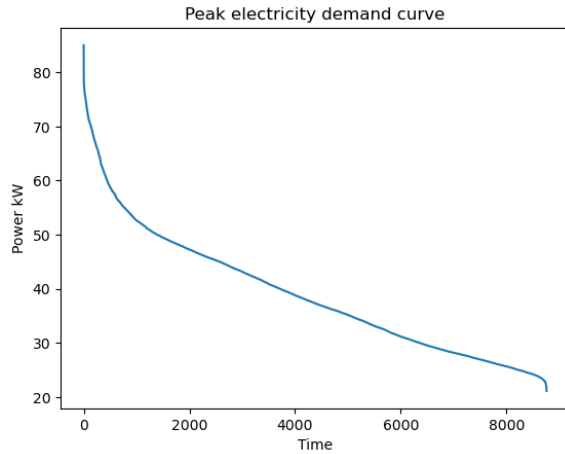
Figure 1: Energy System Diagram

2 Demand Identification and Component Sizing

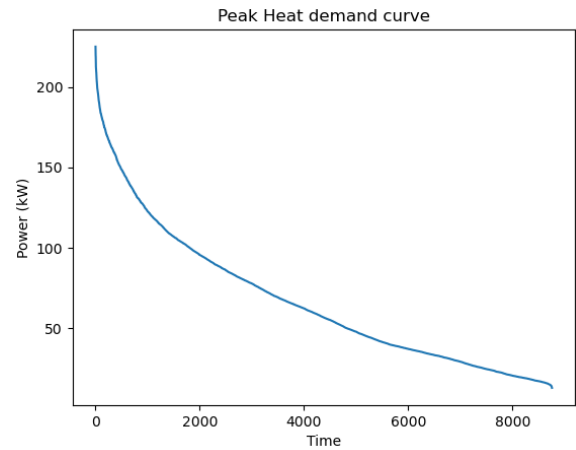
For selection of the energy system components, it is essential to select the component as per the demand of the electricity and heat. Higher size of component will lead to cost burden result into the higher LCOE of the system and less utilisation of the resources. Lower component size is not sufficient to meet the demand leads to the consumer dissatisfaction. Also, sizing the component equal to peak demand is not ideal as that might not last mostly throughout the year. It better to choose two separate components in those cases, one which meets average annual demand operating most of the time of year while second will only operates when demand exceeds the capacity of former component running only for the limited time throughout the year.

To avoid these phenomena, we have analyzed the demand for electricity and heat for the year provided by the Professor. Initial Electricity and Heat demands are given as a nominal value with peak Power demand of 85 kW and peak heat demand of 225 kW. Initially we have obtained the normalized values for both and from that we have calculated

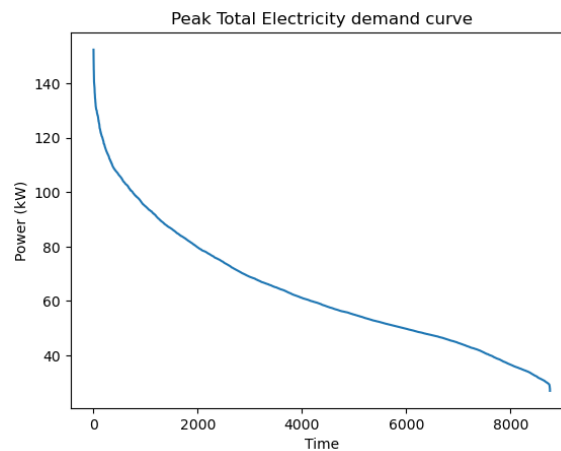
the actual demand for Electricity and Heat. Then, we arranged those values in descending order to generate peak power demand graph, which is given below. From the Peak Heat demand curve, we have finalized the size of heat pump and boiler which is 80kW and 145 kW. Additionally, we assumed that our heat pump will operate around 6000 hours in a year and from the graph we have calculated the excess heat generated during this time which is around 60000 kWh and based on that we decided the size of thermal storage which is 60000 kWh.



Graph 1: Peak Electricity Demand Curve



Graph 2: Peak Heat Demand Curve



Graph 3: Peak Total Electricity Demand Curve

For the sizing of Battery storage and PtoG seasonal storage, the electricity demand for Heat Pump needs to be considered. In our case, we have considered ground water heat pump supplying hot water at 55°C and having source at 0°C. The COP of ground water heat pump in this temperature is around 2.7 [2]. According to that electricity demand for heat pump has been calculated and added to the electricity demand by the consumer. Then again peak demand curve has been generated which is given below. From the graph we find out the peak power demand for the total electricity is around 144kW but that is not actual peak demand as we are not using the heat pump peak heat demand. So, we have added the electricity needed to operate the Heat Pump of capacity 80 kW into the peak electricity demanded by consumer and from that we have calculated the peak

total electricity demand which is around 116 kW. Based on this, we have considered the capacity of fuel cell which is 120kW.

Sizing of the battery storage has been selected by analyzing the daily electricity demand from the users and for the backup of a one single day and based on that it has kept of the capacity 1000 kWh. For the seasonal storage capacity, we calculated the total electricity consumption from the month November till February. Total energy required during this period is around 220 MWh. The fuel cell efficiency and electrolyzer efficiency has been referred from the literature and those are 43% and 53.33% respectively [3]. Based on this calculation the amount of Hydrogen needed to be produced is 15370 kg considering the amount of energy produced by hydrogen per kg is 33.33 kWh. From this, we get to know the size of electrolyzer that needed to produce this amount of Hydrogen within the summer period and that is 120 kW. In PtoG system, we have excluded the water purification process, while gas loss during hydrogen washing process has been considered as charging loss of Hydrogen battery. Also, the sizing of the hydrogen storage and electrolyzer are flexible now, which can be finalized after the optimization of the system.

The capacity of PV system is calculated based on possible installable area, considering the shading effect and necessary gap between the panels when mounted on ground. Currently based on those calculations, PV capacity has been set to the value of 1470 kW. The excess electricity generated by the PV system is feed into the grid. While no energy supply from the grid for the complete energy system.

3 System Components and their Specification

As we have sized our system components, now we need to find their specification for accurate simulation of the system and its Optimization using python. For that we have reviewed the datasheets to get the information regarding the specification of those components, available on the official portal of different manufacturers of same component. Here in the table, we have provided all the information needed for energy system simulation excluding its cost which we will discuss later.

PV modules have been taken by the provider JA Solar with Polycrystalline module having peak load capacity of 450W and around 2m² area needed per module [4]. From the available rooftop area and land area which are 1200m² (facing 30% east, 40% south and 30% west) and 0.75 ha, we have calculated total number of modules possible to install considering the necessary space needed to be provided between installation of solar modules to avoid shading impact and allowing movement between modules for cleaning. As a result, we got to know the possibly installed capacity of 1470kW by installing 3267 modules across the available area. Based on that we have limited our inverter size to the capacity of 1500 kW. We have selected Inverter from Solis company for a better performance and lower cost among the others available in the market [5].

Electricity bus needs to be connected with Battery storage, PtoG seasonal storage, Electricity sink and Heat Pump. Battery storage has been designed for the daily power backup. We have analysed the electricity demand and taken average energy needed to be

stored in battery. As a result, we found the required capacity of Battery to the value 1000 kWh. We have selected Polytech Force H3 model as our base better module. Manufacturer provides stackable battery module with 35.84kWh capacity, Charging/Discharging power of 5.12kW, up to 95% of Depth of Discharge and 0.003% of self-discharge rate per hour [6]. For our system, we need 28 battery modules, combining all providing the net power capacity of 1003.52 kWh and Charging/Discharging power up to 143 kW.

The electrolyzer, hydrogen storage and Fuel cell unit provider for us is H₂ Core System GmbH. In Seasonal storage Water purifier, Battery module and Demand response sensors are considered into our simulation. Based on our energy demand curve, initially we finalised our fuel cell which can fulfill our electricity demand and that is 120 kW from the diagram. So, we kept the size of fuel cell up to value 120 kW with the efficiency of 43% [3]. Based on this we have selected the electrolyzer capacity of 120 kW with efficiency of 53.33% and Hydrogen storage capacity is kept flexible until we get to know the amount of Hydrogen produced by the system after optimising the system. Amount of hydrogen loss during hydrogen drying process considered into hydrogen storage charging efficiency, while no hydrogen compression process has been taken means hydrogen being stored at 35 bar pressure.

Heat pump has been taken with the heating capacity of 80kW. COP for the ground water heat pump has been considered as 2.7 as stated above [2]. Based on assumption of 6000 operating hours of Heat pump, the size of Heat storage has been designed which can functions as daily as well as seasonal heat storage. From the heat peak demand curve and Heat pumps operating hours, the unutilized heat is around 60000 kWh based on which we set the size of heat storage to the 60000 kWh. We are storing heat in the form of sensible heat and therefore we have selected sand battery storage produced by Polar Night Energy with flexible charging and discharging power varying with the change in the flow rate of fluid into the heat exchanger and charging/discharging efficiency of 90% with 0.013% of self-discharge rate per hour. Now, as we are having 225 kW of peak heat demand and 80kW heat capacity of heat pump, we need to fulfill this demand gap by using another heat converter into the system. We have selected the natural gas boiler with power capacity 150 kW and efficiency of 94% which is produced by Bosch company to fill the gap between peak heat demand and demand fulfill by the heat pump [7].

As we have designed all the energy system component needed for the simulation, we also need to consider the break down. For that reason, we have selected a diesel genset with the capacity of 120kW and efficiency of 45% supplied by the Cummins as our back-up energy generation system [8].

4 Simulation and Optimization of Energy System using Python

The microgrid was modeled and simulated in Python. The goal of the implementation was to represent the electricity, heat, and hydrogen domains in one consistent Optimization model and to compute an hourly operating schedule for an entire year that fulfills the given electricity and heat demand without importing electricity from the external grid.

4.1 Modeling Approach

To keep the model transparent and solvable for a full year with hourly resolution, the system was formulated as a linear optimization problem. Each component is represented by energy flows between buses, and the optimization decides the dispatch of converters and storages at every timestep. The main modeling idea is that every demand must be met in every hour, while the available PV generation varies over time. Excess PV can either be stored (battery, hydrogen, thermal storage) or exported to the grid to avoid curtailment.

In the code, the system is built using the oemof.solph framework, which allows defining an energy system as a directed graph: buses represent energy carriers, components create, convert, consume, or store energy, and flows connect them with technical limits and efficiencies. This matches the system concept, where PV is the primary source, battery is used for daily balancing, power to gas provides seasonal electricity storage, and the heat sector is supplied by a heat pump plus a gas boiler for peak demand.

4.2 Input Data Preparation

The optimization is driven by hourly time series over one full year, imported from the project dataset `exam_project_profiles.csv`. The electricity demand is taken from `Pel_demand` (kW) and the heat demand from `Pth_demand` (kW). PV availability is provided as a single aggregated site profile `P_solar_total` (kW), which represents the total PV potential used in the model.

To ensure temporal consistency across all sectors, the dataset is mapped to a uniform hourly time index (starting at 2021-01-01 00:00 with a length equal to the number of rows in the dataset). Prior to constructing the Optimization model, basic input validation is performed: the presence of the required columns is checked, and negative values are clipped to zero. This preprocessing step prevents nonphysical inputs (negative demand or generation) and improves solver robustness for the annual Optimization horizon.

For the PV source in oemof.solph, the aggregated PV potential series is converted into a per-unit availability profile by dividing by the installed PV peak capacity, which allows the installed PV size to remain a configurable design parameter while keeping the temporal shape of the measured or provided PV potential. The resulting per-unit profile is then applied as a fixed generation pattern within the PV source component.

4.3 Energy System Representation in Python

The implemented model follows the structure of the energy flow diagram:

Buses: Separate buses were used for electricity, heat, hydrogen, gas, diesel, and an internal PV DC bus, so that conversions such as DC to AC (inverter) and electricity to hydrogen (electrolyzer) are explicitly represented.

Sources and sinks: PV was added as a renewable source connected to the PV DC bus. Electricity and heat demands were modeled as sinks that must be satisfied in every

timestep. Grid export was modeled as an additional sink with a defined export limit and a revenue term based on the feed in tariff.

Converters: Key converters include the inverter (PV DC to electricity AC), the heat pump (electricity to heat with COP based conversion), the gas boiler (gas to heat), the electrolyzer (electricity to hydrogen), and the fuel cell (hydrogen to electricity). The diesel genset was included as a backup electricity supply, but the Optimization outcome shows that it is not used in the final dispatch under the given assumptions.

Storages: Battery storage was implemented as a short-term electricity buffer, hydrogen storage as seasonal electricity storage coupled with electrolyzer and fuel cell, and thermal storage as daily and seasonal heat storage. The storage models include charging and discharging power limits, round trip efficiencies, self-discharge losses, and initial state of charge.

4.4 Objective Function and Optimization Logic

The Optimization minimises total operational cost over the year while meeting all demands and respecting component constraints. Fuel costs for diesel and natural gas are included as variable costs on the respective input flows. Exported electricity is treated as negative cost (revenue) using the EEG feed in tariff assumption described in the economic section. This setup ensures that PV is used first, then storage charging is scheduled when beneficial, and gas boiler operation is limited to periods where the heat pump plus thermal storage cannot fully cover demand.

A practical addition we included at the modeling level is a seasonal feasibility requirement for the hydrogen storage state of charge around mid-October. This type of constraint helps avoid solutions where hydrogen is depleted too early and improves seasonal reliability for winter supply, which is consistent with the intention of power to gas as seasonal storage.

The optimization problem was solved using the CBC solver through the solph model interface, and afterwards the results were post processed in Python to compute operating hours, energy balances, state of charge curves, and the values required for the LCOE calculation.

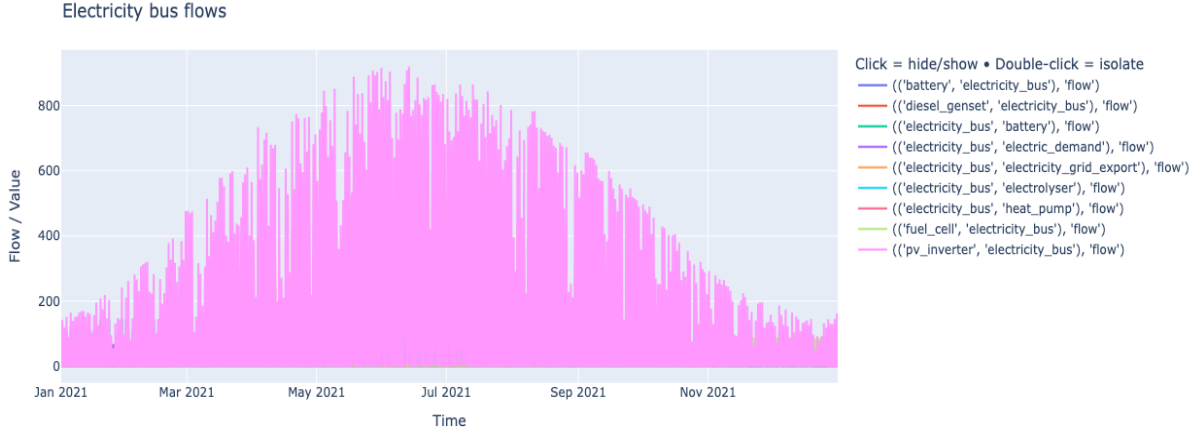
4.5 Post Processing and Validation Checks

After solving, the hourly flow results were aggregated to annual energy values, operating hours were computed from non-zero dispatch periods, and storage state of charge trajectories were extracted for battery, hydrogen, and thermal storage. These outputs are used directly in the results section for interpreting PV production and export, electrolyzer and fuel cell utilization, and the role of thermal storage in shifting heat pump operation.

To ensure the model is consistent, several sanity checks were applied in Python, including verification that electricity and heat demands are met at every timestep, that flows respect nominal limits, and that storage state of charge remains within capacity bounds.

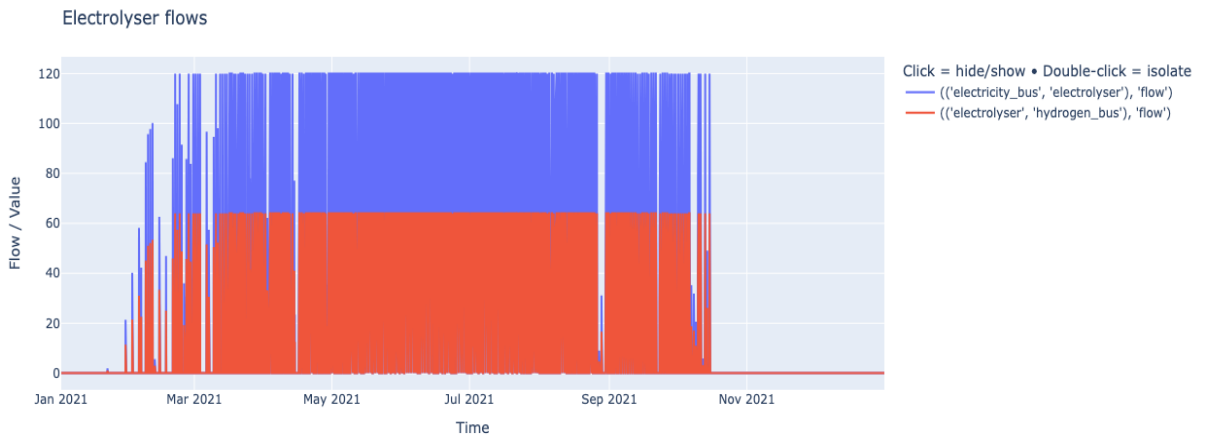
5 Result and Analysis

After optimising energy system, we got to know the total energy produced by the PV modules is 1187507 kWh with the maximum DC power output of 937kW [9]. Also, we are exporting 366370 kWh of solar energy into the grid to avoid the PV curtailment. Export to grid also leads to reduction in LCOE of the system by generating extra income for the grid operator. Now from 937 kW, we are getting 920 kW of maximum PV AC power output from the inverter. The energy generation and consumption chart for the system can be observed as below:

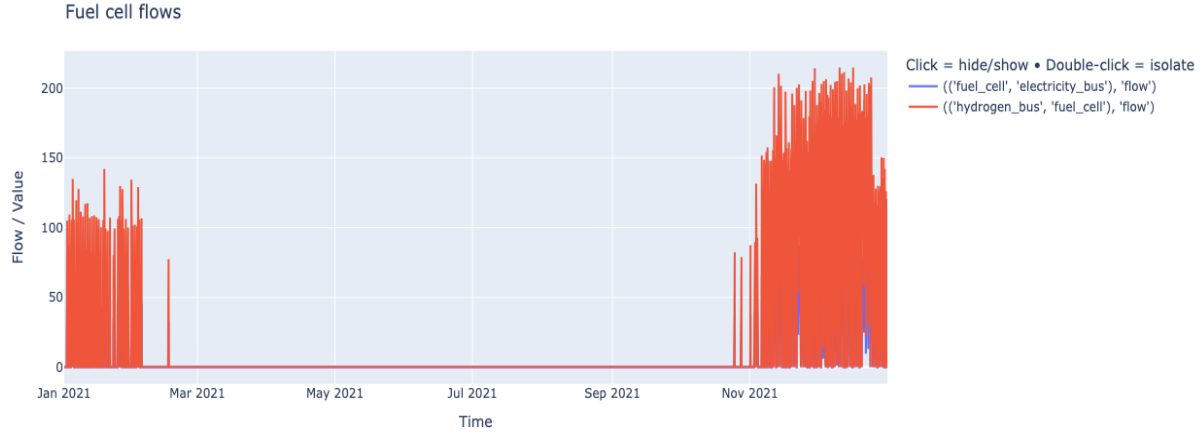


Graph 4: Electricity generation vs consumption

The operating hours for the electrolyzer and fuel cell are 3091 and 1461 respectively, filling hydrogen storage maximum up to 165000 kWh of Hydrogen energy or around 5000 kg of Hydrogen. The flow of seasonal storage is as described in Graph 5 and Graph 6. Chart of SOC for Hydrogen storage and Battery storage can be found in Appedix. Also, there is zero operating hours for the Diesel genset throughout the year.

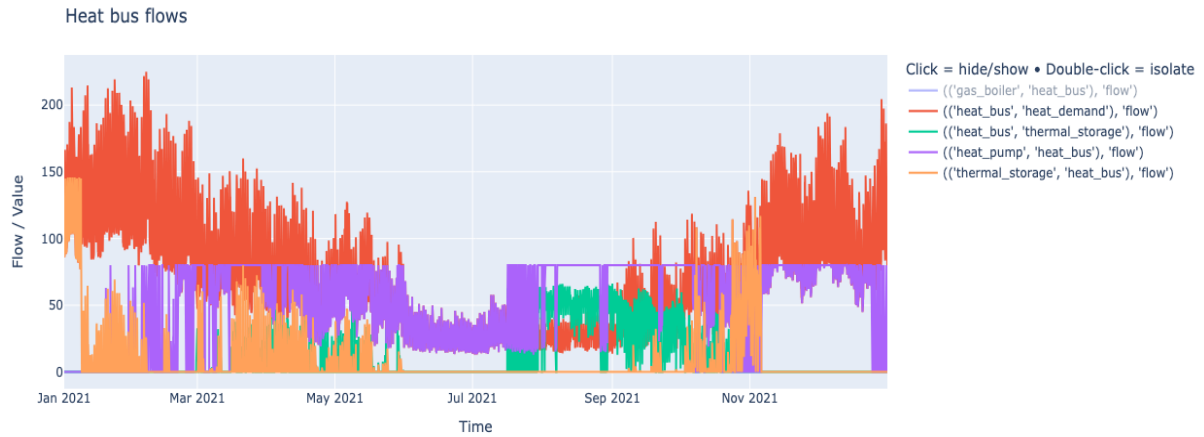


Graph 5: Flow diagram of Electricity and Hydrogen gas in electrolyzer



Graph 6: Flow diagram of Hydrogen gas and Electricity produced in Fuel cell

The operating hours for the heat pump and natural gas boiler are observed as 7137 hours and 2370 hours respectively. The operating hours of natural gas boiler can also be reduced by increasing the size of heat pump. That will also reduce the operating hours of heat pump but consequently increased the capex for the energy system result into the higher LCOE for the system. For that reason, the size of heat pump kept 80kW as described before. The SOC for the thermal storage can be observed in appendix B (Appendix B: Additional Figures), which states that the storage capacity utilised completely, while it can function as daily and seasonal storage. During high solar energy generation, heat pump runs continuously and charges thermal storage, which operates to fulfill heat demand in absence of electricity generation, which can be observed in heat bus flow diagram below:



Graph 7: Heat bus flow diagram

After analyzing energy system optimization result, we got to know that maximum DC power generated by the PV module is around 937 kW because of which we lowered down the inverter capacity to 950 kW. Also, we got to know that the maximum amount hydrogen stored in hydrogen storage is around 5000 kg only, because of which we kept our hydrogen storage volume 5100 kg which can be change whenever needed in the future. Also, we found out that, more amount of solar energy can still be stored by the raise in the battery capacity and electrolyzer capacity. But both are expensive investment and

raises LCOE of the energy system. So, we kept our system designed with the mentioned sizes only.

6 Economic Calculation

For commissioning of the project, consideration of investment and maintenance is essential. It will then give us the levelized cost of the energy, below which rate sell of energy leads us to loss in the project. For calculation of the LCOE below factors has been considered:

- The investment cost of the system apparatus searched on the internet and based on those prices has been considered. The cost for natural gas and diesel is 0.0822 €/kWh and 0.12 €/kWh considered.
- Feed in tariff has been considered as a CfD contract with government, to have constant feed in rate till the solar modules are in operation. According to the EEG, there is no grid maintenance charge for electricity feed in to the grid generated by solar energy. The feed in tariff under EEG for commercial system is 6.72 cents/kWh which has been taken into our calculations [9].
- Also, subsidies available for the particular energy component under various government schemes has been considered. From which net investment (K_n) has been calculated by deducting applicable subsidies from the investment cost.
- The life time of the investment has been considered as mentioned by the manufacture in their data sheets.
- The annual interest rate for our energy system has been considered as 6% by assuming that 50% amount investment has been financed by the KfW scheme.
- The annuity factor for each component has been calculated using the equation,

$$\text{Annuity factor } (A_0) = \frac{\left(\frac{i}{100}\right) \left(1 + \frac{i}{100}\right)^n}{\left(1 + \frac{i}{100}\right)^n - 1} \quad (1)$$

Where, i = Annual interest rate = 6% (for all cases)

n = Lifespan of the component

- Based on this equation, annualised CAPEX value has been calculated as below:

$$\text{CAPEX}(K_0) = \text{Net Investment } (K_n) \times \text{Annuity factor } (A_0) \quad (2)$$

- In this project, we have excluded the calculation of rate of return as no specific price for the cost of energy has been mentioned from the consumer data.
- Maintenance charge has been considered as 2% of the investment cost.
- PV installation cost for roof mounted is 800 €/kW while that is for ground mounted is 700 €/kW taken and feed in to the simulation of the energy system, while in the table below average value has been mentioned.

- Detailed fixed cost can be found in the table below:

Component	Unit cost	Lifespan (years)	Total investment (€)	Subsidy	After subsidy (€)
PV modules	664 €/kW	25	976080	0	976080
Inverter	47 €/kW	10	44650	0	44650
Battery storage	200 €/kWh	15	200000	0	200000
Electrolyzer	2290 €/kW	20	274800	0	274800
Fuel cell	1836.5 €/kW	20	183650	0	183650
Hydrogen storage	0.451 €/kWh	20	74415	0	74415
Heat pump	4000 €/kW	25	320000	55% KfW	144000
Gas boiler	74 €/kW	20	11100	0	11100
Thermal storage	8 €/kWh	25	480000	55% KfW + 2500€ (Häuser Programm, Bavaria)	213500
Diesel genset	–	25	11000	0	11000
Total capital investment					2133195

Table 2: Capital investment cost

- The annualised CAPEX value, Yearly maintenance amount and Revenue generated from the energy exported to the grid are:

Component	Annualised capex (€)
PV modules	102357.63
Inverter	6066.5
Battery storage	20592.55
electrolyzer	23958.32
Fuel cell	16011.44
Hydrogen storage	6684.44
Heat Pump	11264.65
Gas Boiler	967.75
Thermal storage	16701.4
Diesel genset	860.49
Maintenance cost	49356.88
Fuel cost	15470.7
Export Grid maintenance cost	0
Grid revenue	-26692.38
Total	243600.37

Table 3: Annual net cost of the energy model

After providing all these values to the python code, we got our final LCOE for complete project is 26.29 Cents/kWh or 262.9 €/MWh by combining electricity and heat demand. If we consider the energy model designed for only electricity or heat individually then the LCOE in both cases are 63.91 cents/kWh or 639.1 €/MWh for electricity and 8.86 cents/kWh or 88.6 €/MWh for heating system. Which can further be reduced by decreasing PV capacity, but that will increase the use of natural gas boiler and Diesel genset which ends up with higher value of CO₂ emission.

7 Ecological comparison with Conventional Energy Source

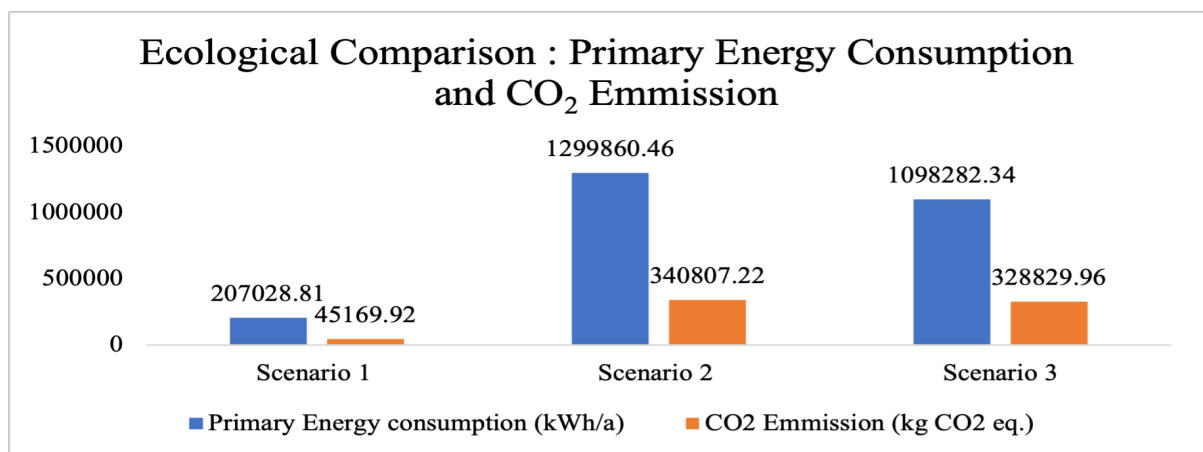
Ecological comparison has been carried out for the energy system and it was compared with basic energy source scenario in which the supply for the electricity demand considered from the electricity grid, while heat demand is getting fulfilled by the natural gas boiler with the efficiency of 94%. The sizing for natural gas boiler has not been performed for conventional energy source scenario, instead it has been calculated based on the energy needed to be supplied to natural gas boiler based on their efficiency. Total Electricity demand and Heat demand from the consumers are 341964.31 kWh/a and 584786.56 kWh/a respectively. The primary energy factor and factor for CO₂ emission has been referred from the standard weighting factors for fuel and electricity according to the German Building Energy Act.

In energy system simulated by us, we are providing electricity from the PV system and diesel genset is not operating. For heat supply, we are supplying 176915.53 kWh/a energy from the natural gas boiler, while rest demand is fulfilled by the Heat pump which also operating on electricity from the renewable energy source. Therefore, the energy needed to supplied to the boiler is 188208.01 kWh/a. As a result, the primary energy consumption for the energy system is 207028.81 kWh/a and CO₂ emitted by the system is 45169.92 kg CO₂ eq./a.

In comparison to the above scenario, the total energy needed to supply to the natural gas boiler is 622113.36 kWh/a. The primary energy consumption for electricity and heating are 615535.76 kWh/a and 684324.70 kWh/a accordingly. The amount of CO₂ emitted by the former is 191500.01 kg CO₂ eq./a, while by the latter is 149307.21 kg CO₂ eq./a. In total primary energy consumed by the conventional energy system is 1299860.46 kWh/a with annual emission of 340807.22 kg CO₂ eq.

In the third scenario, we have considered the heating system as our energy modeling in which heat pump and electricity demand is met by the supply from the grid and for the peak heat demand natural has boiler has been operated. The operating hours for the natural gas boiler and heat pump will remain same with the application of thermal storage into the system. The heat supplied by the natural gas boiler and heat pump are 169965.23 kWh/a and 452898.79 kWh/a. The total electricity demand (Electricity + Heat pump) for the system is 509704.60 kWh/a. The total primary energy consumption by the natural gas boiler and electricity are 180814.07 kWh/a and 917468.27 kWh/a. The CO₂ emission by the natural gas boiler and electricity are 43395.38 and 285434.58 kg CO₂ eq. As a result, total primary energy consumption by the system is 1098282.34 kWh/a and the CO₂ emitted is 328829.96 kg CO₂ eq./a

After calculating the primary energy consumption and CO₂ emission per year, we observed that our energy model is consuming 84% and 81% less compared to the scenario 2 and scenario 3. While for the CO₂ emission, it is 87% lower than the both scenarios.



Graph 8: Ecological comparison

8 Conclusion

After performing energy modeling and its simulation we have found out that,

- Solar energy has a great potential to fulfill the energy demand arise, but the problem is with its storage. We need to designed the proper size of storage system to meet the energy demand during the absence of the solar energy.
- It is difficult to have a PV energy system with peak power generation as exact as peak energy demand. For that reason, the size of solar energy system considered with higher peak value than the peak energy demand value.
- Battery storage system has a better performance as a daily electricity storage system, as it is providing higher efficiency and better performance along with lower response time.
- Battery storage can be used for seasonal storage also as it provides higher round trip efficiency compared to other energy storage system.
- For the storage system, PtoG seasonal storage with hydrogen gas production leads to the higher investment cost with lower round trip efficiency which is around 22.24%, which means that we need to store more than 4 times energy than the energy demand.
- Heat pump has a better performance compared to the other heating system such as boiler using fuel as natural gas, biomass, biogas or electrical, CCHP Plants, Oil boilers or other converters as it provides better performance compared to the rest. The performance of heat pump is also varied by the change of temperature of sink and/or source, type of heat pump such as ground-water, water-water or water-air heat pump.
- It is better to have a thermal storage for the energy system, for storing excess heat when the heat converters are in operation but demand is not there. It can be used a seasonal or daily storage. No separate storage needed to be installed for separate application.
- By implementation of the energy modeling, we can reduce the amount of CO₂ reduction by almost 87% compared the base scenarios.

9 Recommendation

Here, we have modeled and optimized our system based on one possibility only. But further energy system combination also possible to reach the carbon free energy supply, which has their own advantages and disadvantages. We are suggesting few among many possibilities for the energy system modeling which is as below:

- For further reduction in operating hours of natural gas boiler as it is the only CO₂ emitting converter for our energy system modeling, peak power generation capacity of PV system needed. As a result, it will increase the investment cost which leads to the more LCOE.

- Increase in electrolyzer capacity can also lead to reduction in operating hours of boiler, but as it has a lower round trip efficiency again it will raise the cost of energy system.
- Natural gas boiler can also be replaced with electric boiler or heat pump, but as a result we need to operate diesel genset as fuel cell alone can not fulfill the energy demand during peak energy consumption when PV energy is not there. To meet that energy demand, again we need to raise the capacity of electrolyzer and Fuel cell.
- Boilers operation can again be reduced down by increasing the size of thermal storage, which seems feasible solution as it has lower investment compared to any other storage.
- It better to opt for other seasonal storage than the PtoG by hydrogen gas production, as other storages have better round-trip efficiency and lower investment. It will also help in building energy converters with higher power rating which helps us in more utilization of solar energy. As a suggestion, we would like to prefer battery storage as seasonal storage also as it has around 94% round trip efficiency compared other seasonal storage such Pump storage Hydro Powerplant, PtoG storage or thermal storage.
- Currently, we are exporting excess electricity to grid for avoiding PV curtailment as it's cost free for energy export to grid if generated from the renewable energy source. But in many countries, producers need to pay fees for grid export which is higher than the price of energy they are getting by exporting. In such conditions its better use the inverter with the peak power capacity of our system requirement and curtail the excess energy generated by the PV system.

10 References

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11 Individual Contribution Statement

Team Member Contributions

Student Name: Vasim Husenbhai Chudesara

- Main responsibilities: Design of Energy model, Defining function of each system components, System Sizing, Testing and Validation of Optimization code, Report writing
- Time contribution: 25% of total project effort

Student Name: Ashish Manoj Mistari

- Main responsibilities: Annual PV power generation using PVGIS, Data collection such as components performance parameters and costing for the system component, Finding subsidies and policies related to the system components, Report Writing
- Time contribution: 25% of total project effort

Student Name: JESTEENA JOSE

- Main responsibilities: Simulation and Optimization of Energy model into Python, Solving python queries, Report writing
- Time contribution: 25% of total project effort

Student Name: Jetson Joseph

- Main responsibilities: Simulation and Optimization of Energy model into Python, Solving python queries, Report writing
- Time contribution: 25% of total project effort

Collaborative Work

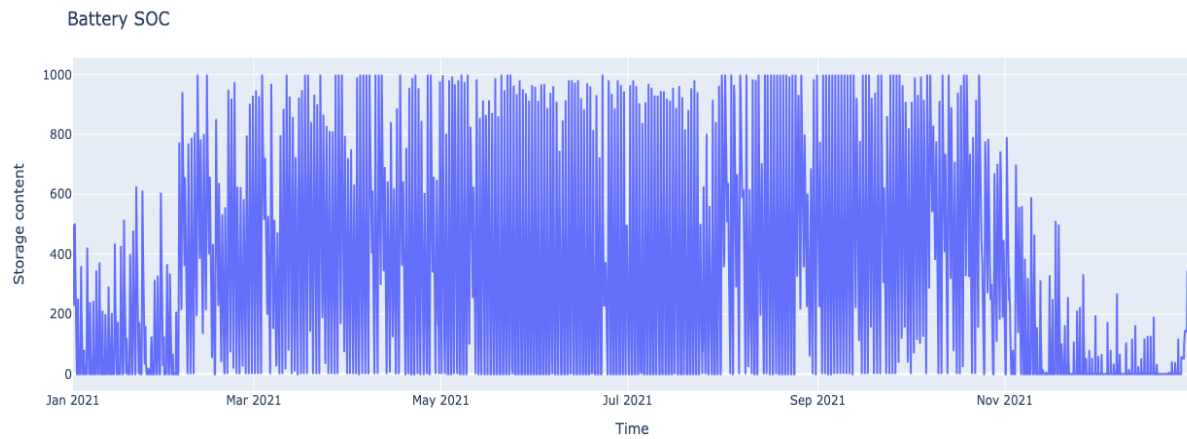
From the beginning of the project, we all are working as a team. Indeed, we have divided our task individually, but every thought and information have been verified and accepted in a team meeting and then only implemented into the project. This project is a result of our collaborative work, rather than the individual tasks.

A Appendix A : Project Files

Project GitLab Link

B Appendix B: Additional Figures

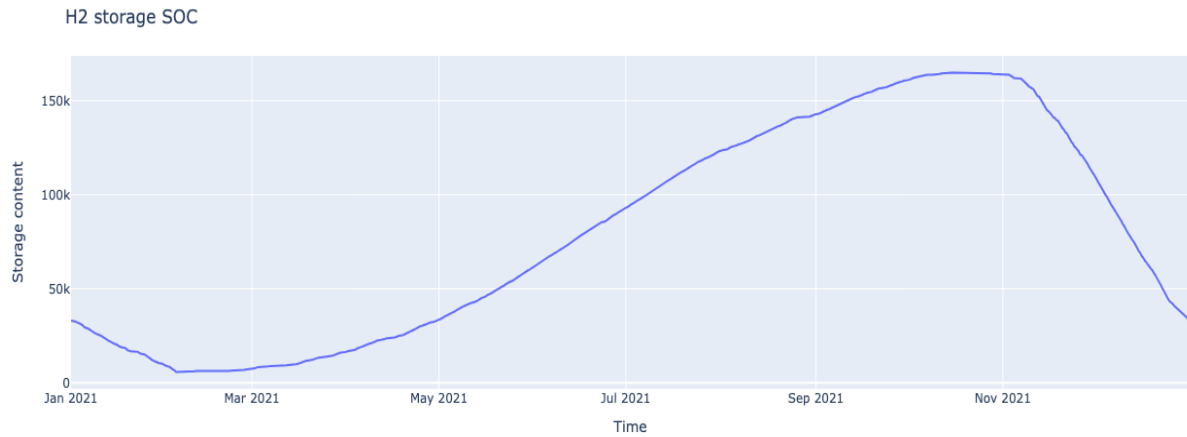
- Annual SOC graph for Battery storage, Hydrogen Storage and Thermal Storage and state of Diesel genset



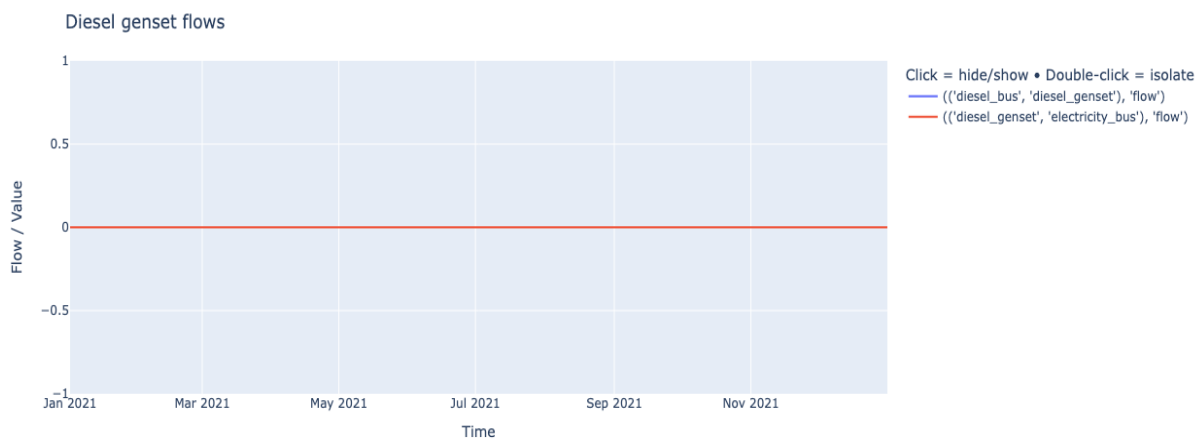
Graph 9: Annual SOC curve for battery storage



Graph 10: Annual SOC curve for Thermal storage

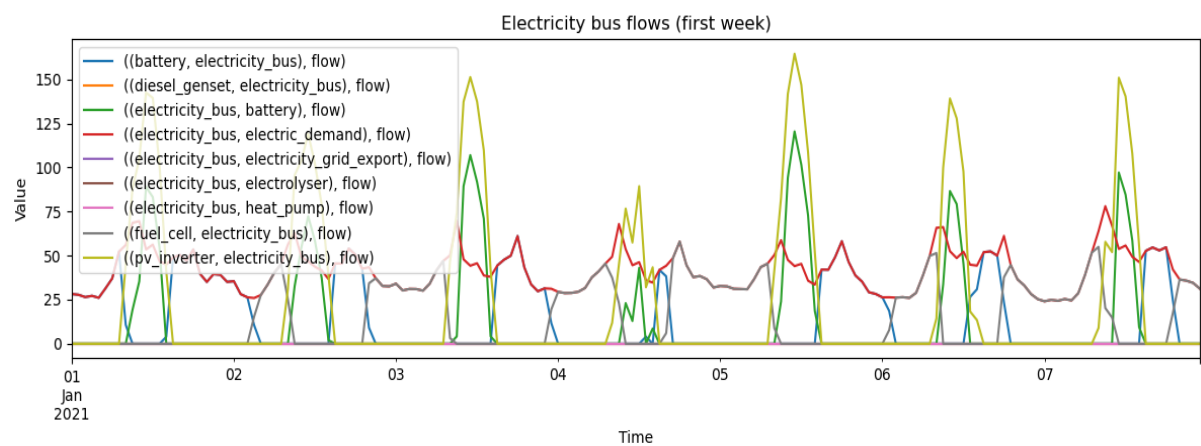


Graph 11: Annual SOC curve for Hydrogen storage

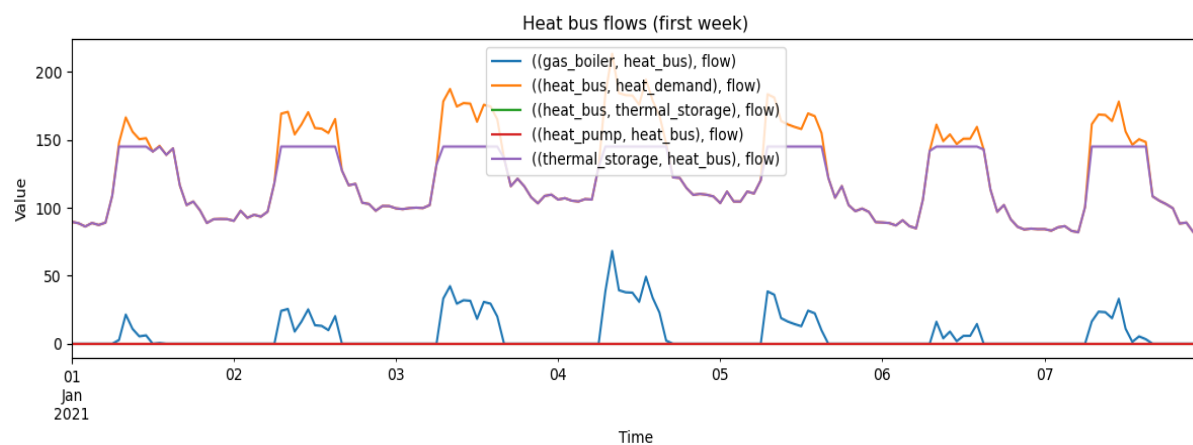


Graph 12: State of Diesel genset

- System flow diagram for the first week for electricity and Heat



Graph 13: Electricity bus flow diagram for the first week



Graph 14: Heat bus flow diagram for the first week